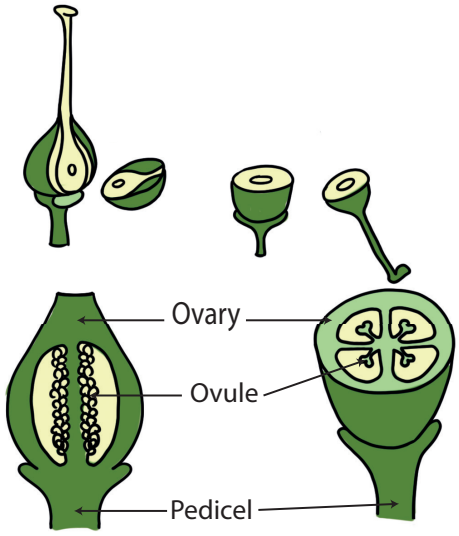
Cut a ovary of a flower both vertically and horizontally. Observe the ovules. Compare the ovary and ovules from few different flowers. Are there one or more ovules? Can you see any connection between the number of ovules in the ovary and number of seeds in each fruit?



Longitudinal Section of an Ovary **Transverse Section of an Ovary**

Collect some fruits like tomato, brinjal, lady's finger (vegetable), mango, peas and custard apple and observe. You can see some green part above brinjal and lady's finger. What are they?

Compare mango, custard apple and peas. All these are single fruits but custard apple has many small parts in it, each with a seed. Mango has a single seed and pea has many seeds. What do you understand from the above observations?

- A green part above fruits of **brinjal** and **lady's finger** are sepals of a flower. In some plants, after fertilization, sepal will not fall from fruit and remain or persist with fruit.
- **Custard apple** is made up of many fruits, aggregated together. Each fruit part is thin, membranous with some granule like, which is edible.
- In **mango** the outer skin and middle pulpy are edible and sweet. The inner most part is with single seed.
- In **pea** the fruit is not fleshy, but forms a covering pouch for many seeds.

In all the above fruits, ovary, a lower most swollen part of pistil develops into a fleshy fruit. Ovules present inside the ovary gets transformed into a seed.

Hence, now with these observations, we shall list the changes taking place in a flower after fertilization. These are collectively said to be **post fertilization changes**

- ❖ Calyx sometimes persist with fruit.
- ❖ Petals fall off.
- ❖ Androecium fall off.
- ❖ Pistil remain and develops into a fruit.
- ❖ Style and stigma fall off
- ❖ Ovary enlarges to store food materials and develops into a fruit.
- ❖ Ovules present inside the ovary develops into seeds.



The world's largest and heaviest seed is the double coconut. The seed looks like two coconut fused together. It grows only in two islands of the Seychelles. A single seed may be 12 inches long, nearly 3 feet in circumference and weighs about 18 kg.

Orchids have the smallest seeds in the plant kingdom. 35 million seeds may weight only about 25 gram.